GRADE 12 CHEMISTRY

UNIT 3 — INDUSTRIAL CHEMISTRY

Raw materials, chemical manufacture and manufacturing industries in Ethiopia

Exam-focused revision booklet

Prepared from the uploaded Ethiopian Grade 12 Chemistry Student Textbook (Ministry of Education, 2023). The explanations below are newly written study notes, not copied textbook pages. Diagrams are original schematic summaries.

How to use this booklet

Read the core idea, memorize the key relationship or equation, redraw the visual summary from memory, then answer both alternative questions without looking. For calculations, show units and working. For process questions, give the stages in the correct order.

CORE CONCEPTS	KEY EQUATIONS / PROCESSES	EXAM PRACTICE

Coverage: 19 focused topic sheets plus this guide page = 20 pages minimum. The two questions on each topic are alternatives for self-testing.

3.01 Industrial chemistry and society

Core concept

Industrial chemistry applies chemical knowledge to transform raw materials into useful products on a large scale. Chemical industries supply materials such as fertilizers, acids, plastics, fibers, dyes and other products. The textbook emphasizes their importance for economic growth and development, including Ethiopia's need to expand domestic production.

Remember

Raw materials → processing → products + by-products

Industrial chemistry and society	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. Define industrial chemistry and state two roles in society.

B. Give five examples of products supplied by chemical industries.

3.01 • Exam diagram & MCQ practice

Key exam idea: Industrial chemistry applies chemical knowledge to manufacture useful products on a large scale from raw materials.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. Which sequence best represents a basic industrial process?

A. raw materials → processing → products
B. products → raw materials → mining
C. waste → catalyst → ore
D. energy → products → raw materials

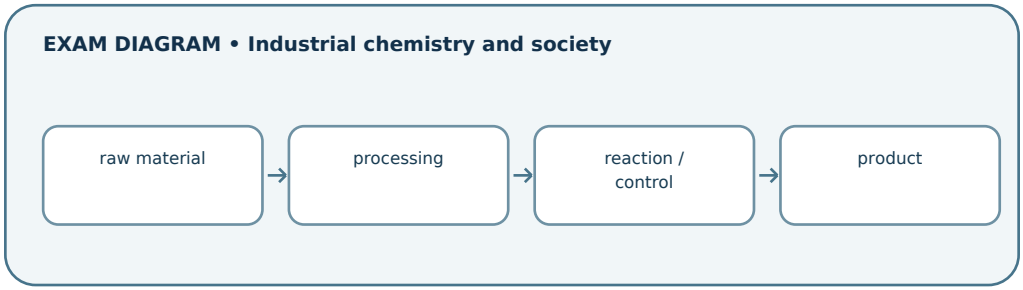
Answer: A. Manufacturing begins with feedstocks, processes them, and produces useful products and sometimes by-products.

Question 2. In the diagram, which labelled step comes immediately after “raw material”?

A. processing
B. reaction / control
C. product
D. raw material

Answer: A. The schematic is arranged in sequence: raw material → processing → reaction / control → product.

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.



3.02 Natural resources as raw materials

Core concept

Raw materials are natural resources used as inputs in manufacturing. They may be mineral, plant, animal, water or air-derived resources. A resource becomes industrially valuable when it can be processed economically into a useful product. Consider availability, quality, transport, energy, environmental effects and sustainability.

Remember

Resource → extraction/preparation → manufacture

Natural resources as raw materia	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. Distinguish a natural resource from a manufactured product.

B. List examples of Ethiopian mineral resources mentioned in the textbook.

3.02 • Exam diagram & MCQ practice

Key exam idea: Natural resources such as minerals, air, water and biomass can serve as industrial feedstocks.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. Which is a natural resource that can be used as an industrial raw material?

A. limestone
B. polyethylene bag
C. glass bottle
D. synthetic nylon fibre

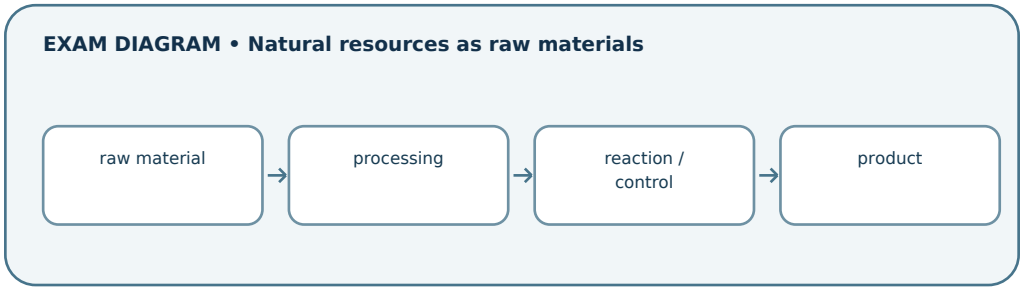
Answer: A. Limestone is a naturally occurring mineral used in cement and other industries.

Question 2. In the diagram, which labelled step comes immediately after “raw material”?

A. processing
B. reaction / control
C. product
D. raw material

Answer: A. The schematic is arranged in sequence: raw material → processing → reaction / control → product.

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.



3.03 Industry and industrial processes

Core concept

An industrial process converts feedstocks into products through physical and chemical operations. Important considerations include yield, selectivity, energy, catalysts, waste treatment, safety and product quality. Chemical equations describe the transformations, while process-flow diagrams show the sequence of operations. Identify inputs, conditions, outputs and recycle streams.

Remember

Feedstock + energy + process → product + by-products

Industry and industrial processe	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. List general characteristics of chemical industries.

B. Why are catalysts and process control important in industrial manufacture?

3.03 • Exam diagram & MCQ practice

Key exam idea: Industrial processes require suitable feedstocks, energy, equipment, process control and often catalysts.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. What is the main function of a catalyst in many industrial reactions?

A. increase reaction rate without being consumed overall
B. increase product mass by creating atoms
C. shift every equilibrium completely to products
D. replace all raw materials

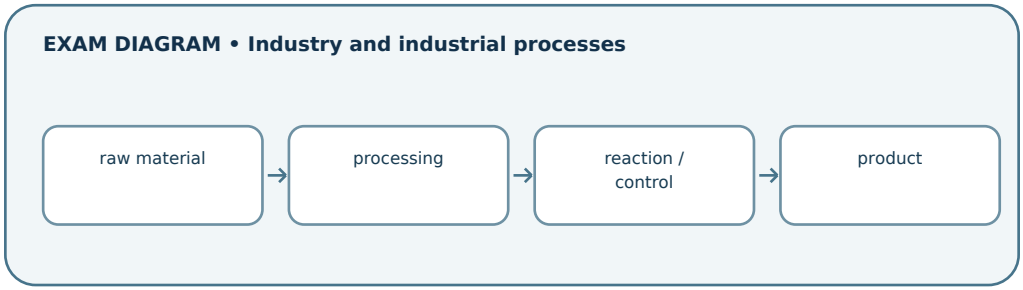
Answer: A. A catalyst provides an alternative pathway with lower activation energy and is regenerated.

Question 2. In the diagram, which labelled step comes immediately after “raw material”?

A. processing
B. reaction / control
C. product
D. raw material

Answer: A. The schematic is arranged in sequence: raw material → processing → reaction / control → product.

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.



3.04 Ammonia manufacture — Haber process

Core concept

The Haber process combines nitrogen and hydrogen to produce ammonia. The reaction is reversible and exothermic. Industrial conditions balance rate and equilibrium yield; a catalyst is used, and unreacted gases are recycled. The textbook discusses feed gases, operating conditions and the importance of ammonia as a starting material for fertilizers and other chemicals.

Remember

N■(g) + 3H■(g) ■ 2NH■(g)

Ammonia manufacture — Haber proc	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. Write the balanced equation for ammonia manufacture.

B. Explain why pressure, temperature and catalyst are important in the Haber process.

3.04 • Exam diagram & MCQ practice

Key exam idea: The Haber process is $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$, an exothermic reversible reaction using an iron catalyst.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. Which is the balanced Haber-process equation?

A. $N_2 + 3H_2 \rightleftharpoons 2NH_3$
B. $N_2 + H_2 \rightleftharpoons NH_3$
C. $2N_2 + 3H_2 \rightleftharpoons 2NH_3$
D. $N_2 + 2H_2 \rightleftharpoons NH_4$

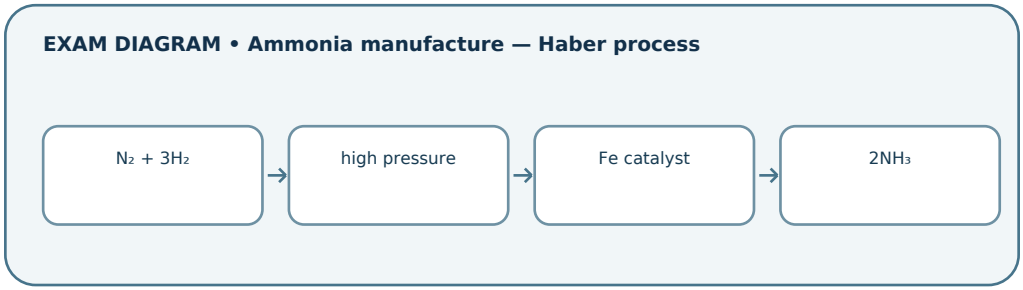
Answer: A. The balanced reaction conserves two nitrogen atoms and six hydrogen atoms on each side.

Question 2. In the diagram, which labelled step comes immediately after "$N_2 + 3H_2$"?

A. high pressure
B. Fe catalyst
C. $2NH_3$
D. $N_2 + 3H_2$

Answer: A. The schematic is arranged in sequence: $N_2 + 3H_2$ → high pressure → Fe catalyst → $2NH_3$.

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.



3.05 Ammonia: equilibrium and uses

Core concept

Because ammonia synthesis is reversible, Le Châtelier’s principle helps explain industrial operating choices. Increasing pressure favors the side with fewer gas molecules; lower temperature favors the exothermic forward reaction at equilibrium, while a catalyst speeds attainment of equilibrium. Ammonia is important in nitrogen fertilizers and chemical manufacture.

Remember

N■ + 3H■ ■ 2NH■; exothermic

Ammonia	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. Predict the effect of increasing pressure on ammonia equilibrium.

B. State two industrial uses of ammonia.

3.05 • Exam diagram & MCQ practice

Key exam idea: Higher pressure favours ammonia formation because the product side has fewer gas moles; lower temperature favours the exothermic direction.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. Increasing pressure in the Haber equilibrium shifts it toward:

A. ammonia
B. nitrogen and hydrogen
C. no side because gas moles are equal
D. oxygen

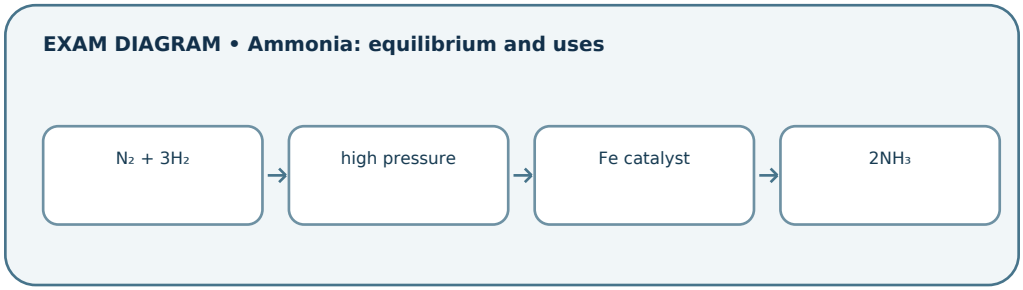
Answer: A. There are four moles of gas on the reactant side and two on the product side, so pressure favours NH_3.

Question 2. In the diagram, which labelled step comes immediately after "$N_2 + 3H_2$"?

A. high pressure
B. Fe catalyst
C. $2NH_3$
D. $N_2 + 3H_2$

Answer: A. The schematic is arranged in sequence: $N_2 + 3H_2$ → high pressure → Fe catalyst → $2NH_3$.

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.



3.06 Nitric acid manufacture

Core concept

The Ostwald process converts ammonia into nitric acid through successive oxidation and absorption stages. Ammonia is oxidized to nitrogen monoxide, which is further oxidized to nitrogen dioxide; the nitrogen dioxide is absorbed in water to produce nitric acid. The sequence and balanced equations are important examination targets.

Remember

NH■ → NO → NO■ → HNO■

Nitric acid manufacture	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. Arrange the main stages of nitric acid manufacture in order.

B. Write the oxidation of NO to NO■ and explain the role of oxygen.

3.06 • Exam diagram & MCQ practice

Key exam idea: The Ostwald process oxidizes NH_3 to NO, then NO to NO_2, followed by absorption in water to form HNO_3.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. Which sequence correctly summarizes the Ostwald process?

A. $NH_3 \rightarrow NO \rightarrow NO_2 \rightarrow HNO_3$
B. $NO_2 \rightarrow NH_3 \rightarrow NO \rightarrow HNO_3$
C. $N_2 \rightarrow H_2 \rightarrow NH_3 \rightarrow HNO_3$
D. $HNO_3 \rightarrow NO_2 \rightarrow NO \rightarrow NH_3$

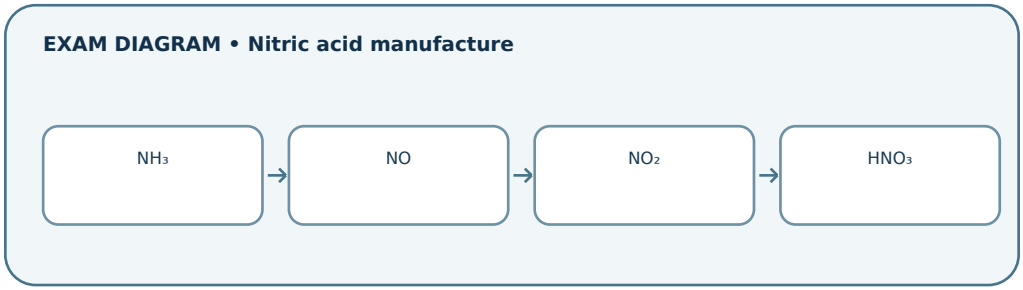
Answer: A. Ammonia is catalytically oxidized to NO, further oxidized to NO_2, and absorbed to produce nitric acid.

Question 2. In the diagram, which labelled step comes immediately after “NH_3”?

A. NO
B. NO_2
C. HNO_3
D. NH_3

Answer: A. The schematic is arranged in sequence: $NH_3 \rightarrow NO \rightarrow NO_2 \rightarrow HNO_3$.

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.



3.07 Nitrogen-based fertilizers

Core concept

Nitrogen fertilizers supply plants with nitrogen needed for growth. The textbook treats examples and production routes linked to ammonia and nitric acid. Compare fertilizers by their nitrogen-containing ions or compounds, nutrient content, use and handling. Excessive or poorly managed fertilizer use can contribute to environmental problems, so application should be appropriate.

Remember

Ammonia is a key feedstock for many N fertilizers

Nitrogen-based fertilizers	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. Name nitrogen-based fertilizers discussed in the unit.

B. Explain the importance of nitrogen fertilizer and one possible environmental concern.

3.07 • Exam diagram & MCQ practice

Key exam idea: Nitrogen fertilizers supply plant-available nitrogen, but excessive application can cause nitrate leaching and eutrophication.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. Which is a possible environmental consequence of excessive nitrogen fertilizer use?

A. eutrophication of water bodies
B. complete prevention of algal growth
C. elimination of soil nutrients
D. formation of pure oxygen

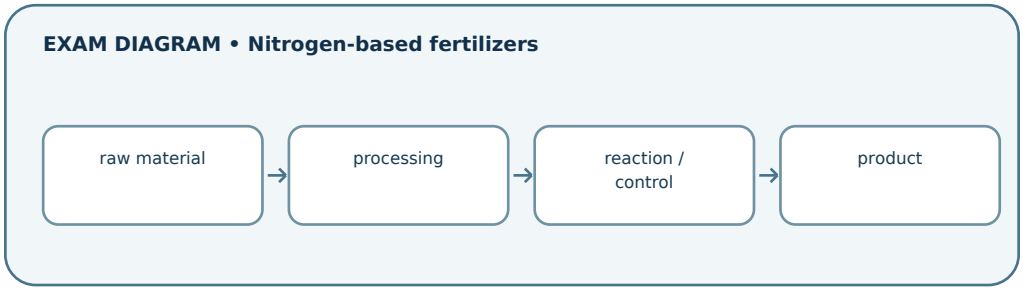
Answer: A. Nitrate runoff can enrich water, stimulate algal blooms and reduce dissolved oxygen during decomposition.

Question 2. In the diagram, which labelled step comes immediately after “raw material”?

A. processing
B. reaction / control
C. product
D. raw material

Answer: A. The schematic is arranged in sequence: raw material → processing → reaction / control → product.

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.



3.08 Sulfuric acid manufacture

Core concept

Sulfuric acid is manufactured industrially by the Contact process. Sulfur or sulfur-containing materials are converted to sulfur dioxide; SO■ is oxidized to SO■ using a catalyst, and SO■ is absorbed to form oleum before dilution. The process is designed to obtain high conversion while controlling heat and avoiding inefficient direct hydration.

Remember

S + O■ → SO■; 2SO■ + O■ ■ 2SO■

Sulfuric acid manufacture	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. Outline the stages of the Contact process.

B. Why is SO■ absorbed in concentrated sulfuric acid rather than simply added directly to water?

3.08 • Exam diagram & MCQ practice

Key exam idea: The Contact process converts sulfur to SO_2, oxidizes SO_2 to SO_3 using V_2O_5, then absorbs SO_3 in concentrated H_2SO_4.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. Which catalyst is commonly used in the Contact process?

A. vanadium(V) oxide
B. iron
C. nickel
D. platinum black only

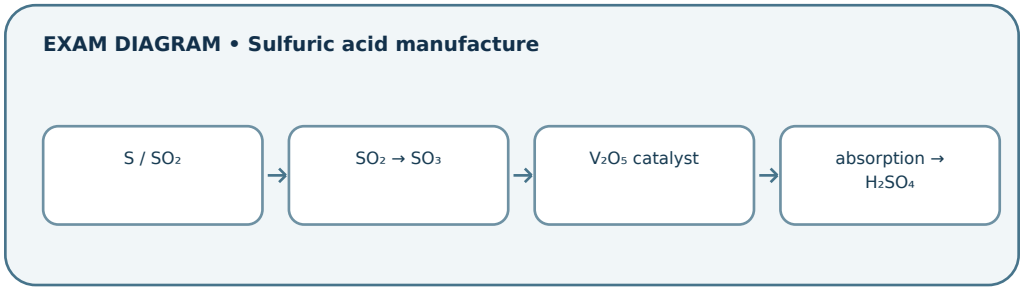
Answer: A. V_2O_5 catalyzes the reversible oxidation of SO_2 to SO_3.

Question 2. In the diagram, which labelled step comes immediately after “S / SO_2”?

A. $SO_2 \rightarrow SO_3$
B. V_2O_5 catalyst
C. absorption → H_2SO_4
D. S / SO_2

Answer: A. The schematic is arranged in sequence: S / SO_2 → SO_2 → SO_3 → V_2O_5 catalyst → absorption → H_2SO_4.

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.



3.09 Sulfuric acid: properties and uses

Core concept

Sulfuric acid is a major industrial chemical used in fertilizer manufacture and other chemical processes. Its concentration and strong dehydrating/corrosive behavior require careful handling. In exam answers, connect its industrial importance to the scale of use and identify the process steps and relevant equations accurately.

Remember

Contact process produces H■SO■

Sulfuric acid	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. Give three uses of sulfuric acid.

B. Explain why industrial sulfuric acid manufacture is important to other industries.

3.09 • Exam diagram & MCQ practice

Key exam idea: Sulfuric acid is a major industrial chemical used in fertilizers, petroleum processing and manufacture of other chemicals.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. Which industry is a major user of sulfuric acid?

A. fertilizer manufacture
B. glass recycling only
C. paper printing only
D. food refrigeration

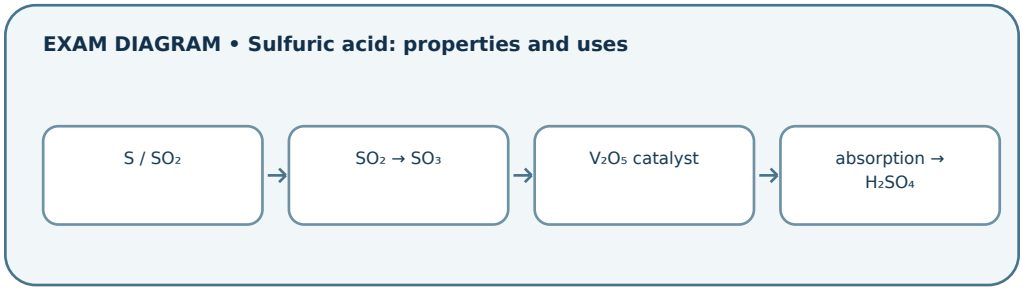
Answer: A. Sulfuric acid is used to manufacture phosphate fertilizers and many other industrial chemicals.

Question 2. In the diagram, which labelled step comes immediately after “S / SO_2”?

A. $SO_2 \rightarrow SO_3$
B. V_2O_5 catalyst
C. absorption → H_2SO_4
D. S / SO_2

Answer: A. The schematic is arranged in sequence: S / SO_2 → SO_2 → SO_3 → V_2O_5 catalyst → absorption → H_2SO_4.

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.



3.10 Pesticides and herbicides

Core concept

Pesticides are substances used to control pests; herbicides specifically control unwanted plants. The textbook introduces common examples and their roles. Their benefits include protecting crops and reducing losses, but inappropriate use may affect non-target organisms, human health and ecosystems. Distinguish the intended target, purpose and possible risk.

Remember

Pesticide = pest control; herbicide = weed control

Pesticides and herbicides	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. Differentiate pesticides from herbicides.

B. State one benefit and one risk of pesticide use.

3.10 • Exam diagram & MCQ practice

Key exam idea: Pesticides control pests; herbicides are a type of pesticide specifically used to control unwanted plants.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. A chemical used specifically to control weeds is a:

A. herbicide
B. fungicide
C. insecticide
D. rodenticide

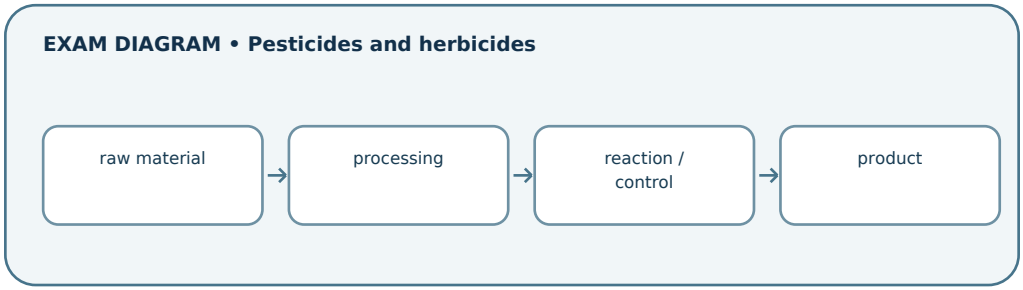
Answer: A. Herbicides target unwanted plants, including weeds.

Question 2. In the diagram, which labelled step comes immediately after “raw material”?

A. processing
B. reaction / control
C. product
D. raw material

Answer: A. The schematic is arranged in sequence: raw material → processing → reaction / control → product.

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.



3.11 Sodium carbonate

Core concept

Sodium carbonate is an important industrial chemical used in products such as glass and other manufacturing processes. The textbook describes its manufacture and applications. When revising, learn the raw materials, sequence of operations, key reactions and how the product is separated. Connect the chemical to its industrial role rather than memorizing only its formula.

Remember

Sodium carbonate = Na■CO■

Sodium carbonate	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. State two uses of sodium carbonate.

B. Describe the main idea of the industrial process for producing sodium carbonate.

3.11 • Exam diagram & MCQ practice

Key exam idea: Sodium carbonate (Na_2CO_3) is used in glass manufacture, detergents and water softening.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. Which is a common use of sodium carbonate?

A. manufacture of glass
B. vulcanization of rubber
C. production of nylon fibres
D. electroplating copper

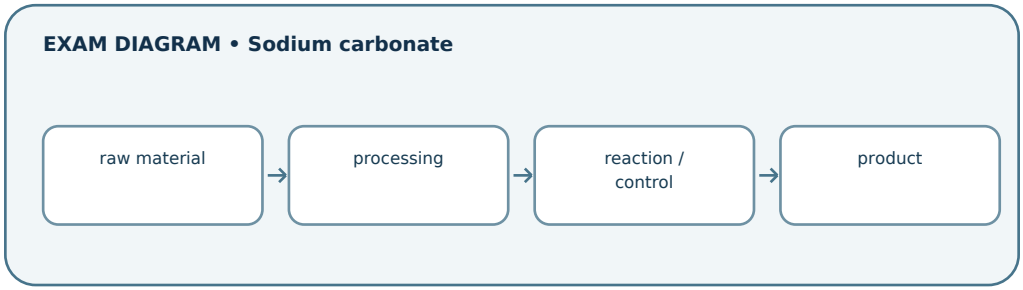
Answer: A. Sodium carbonate is an important raw material in soda-lime glass manufacture.

Question 2. In the diagram, which labelled step comes immediately after “raw material”?

A. processing
B. reaction / control
C. product
D. raw material

Answer: A. The schematic is arranged in sequence: raw material → processing → reaction / control → product.

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.



3.12 Sodium hydroxide

Core concept

Sodium hydroxide (caustic soda) is an important base used in industrial cleaning and manufacture of other products. Its strong alkalinity makes it useful but also hazardous and corrosive. The textbook includes its manufacture and uses. In answers, distinguish the chemical role from safe handling and health effects.

Remember

NaOH; strong base

Sodium hydroxide	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. List industrial uses of sodium hydroxide.

B. Explain why caustic soda requires careful handling.

3.12 • Exam diagram & MCQ practice

Key exam idea: Sodium hydroxide is a strong alkali used in soap, paper, textiles and chemical manufacture.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. Sodium hydroxide is also known as:

A. caustic soda
B. washing soda
C. baking soda
D. quicklime

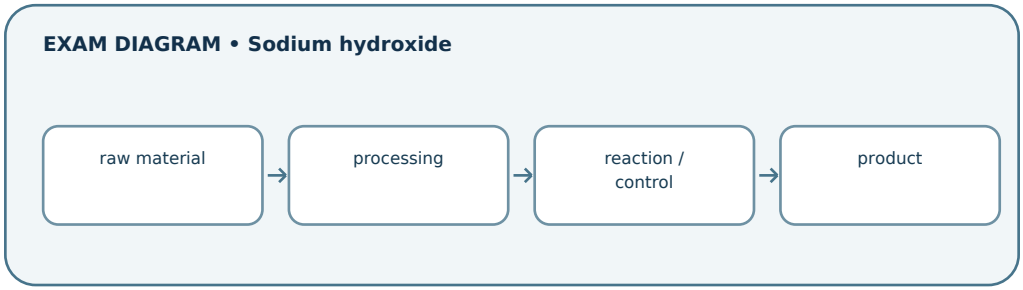
Answer: A. NaOH is commonly called caustic soda; sodium carbonate is washing soda.

Question 2. In the diagram, which labelled step comes immediately after “raw material”?

A. processing
B. reaction / control
C. product
D. raw material

Answer: A. The schematic is arranged in sequence: raw material → processing → reaction / control → product.

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.



3.13 Glass: composition and properties

Core concept

Glass is an amorphous or non-crystalline solid. Silica is its main component. Glass is useful because it can be shaped while molten, is chemically resistant in many uses and can be recycled. Quartz glass is made from very pure silica; soda-lime glass is ordinary glass made from silica, sodium carbonate or sulfate, and limestone.

Remember

Main glass former: SiO■

Glass	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. Define glass and state why its structure is described as amorphous.

B. Name the main raw materials used to make soda-lime glass.

3.13 • Exam diagram & MCQ practice

Key exam idea: Glass is an amorphous inorganic solid; silica is its main structural component, with additives modifying properties.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. The principal raw material in ordinary soda-lime glass is:

A. silica (SiO_2)
B. cellulose
C. calcium carbonate alone
D. sodium chloride

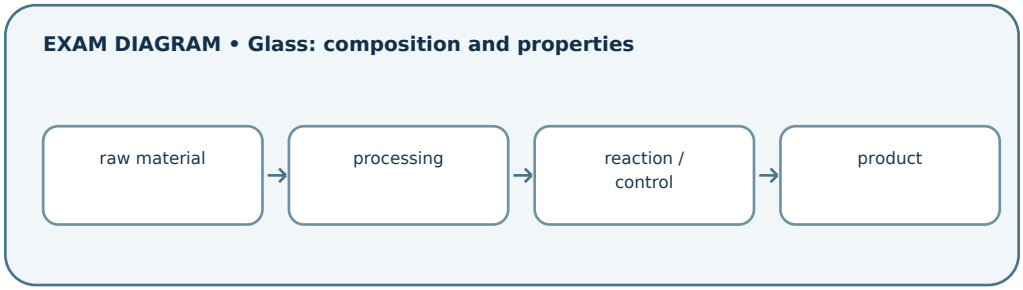
Answer: A. Silica forms the main glass network; soda and lime modify melting and durability.

Question 2. In the diagram, which labelled step comes immediately after “raw material”?

A. processing
B. reaction / control
C. product
D. raw material

Answer: A. The schematic is arranged in sequence: raw material → processing → reaction / control → product.

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.



3.14 Ceramics

Core concept

Ceramics are manufactured from inorganic non-metallic materials, commonly shaped from clay or related raw materials and hardened by firing. The production sequence includes preparing the raw materials, shaping, drying and firing. Properties and uses depend on composition and processing. Compare traditional clay products with manufactured ceramics using their processing and final properties.

Remember

Raw material preparation → shaping → drying → firing

Ceramics	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. Outline the important steps in ceramic manufacture.

B. Give two uses of ceramics and explain one property that makes them suitable.

3.14 • Exam diagram & MCQ practice

Key exam idea: Ceramics are inorganic, non-metallic materials shaped and fired at high temperatures; they are often hard and heat-resistant.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. Which process is essential in conventional ceramic manufacture?

A. firing at high temperature
B. electrolysis of molten clay
C. fermentation
D. polymer chain-growth

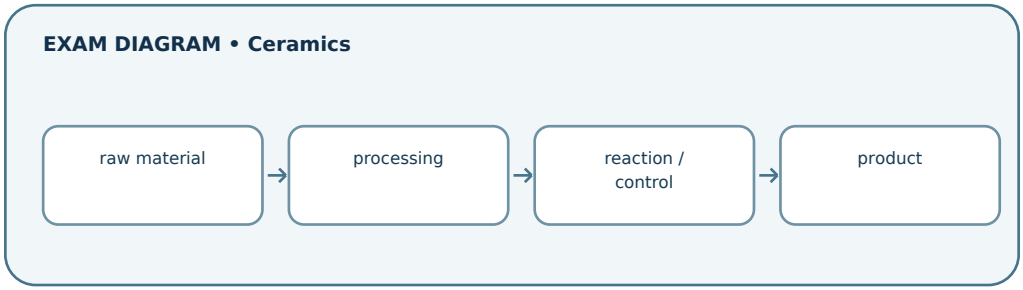
Answer: A. Firing develops strength and the required ceramic structure through heat-driven changes.

Question 2. In the diagram, which labelled step comes immediately after “raw material”?

A. processing
B. reaction / control
C. product
D. raw material

Answer: A. The schematic is arranged in sequence: raw material → processing → reaction / control → product.

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.



3.15 Cement

Core concept

Cement is a binding material produced by processing mineral raw materials at high temperature. Manufacture involves preparing and mixing raw materials, heating to form clinker, cooling, grinding and adding suitable material such as gypsum. Cement reacts with water and hardens, making it essential in construction. Learn the process sequence and role of each stage.

Remember

Raw mix → kiln/clinker → cooling → grinding

Cement	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. Arrange the main stages of cement manufacture.

B. What is the role of gypsum in cement production?

3.15 • Exam diagram & MCQ practice

Key exam idea: Cement is made by heating limestone and clay-containing materials to form clinker, which is ground with gypsum.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. What is the main role of gypsum added during cement grinding?

A. control the setting time
B. provide the main fuel
C. remove all silica
D. turn cement into glass

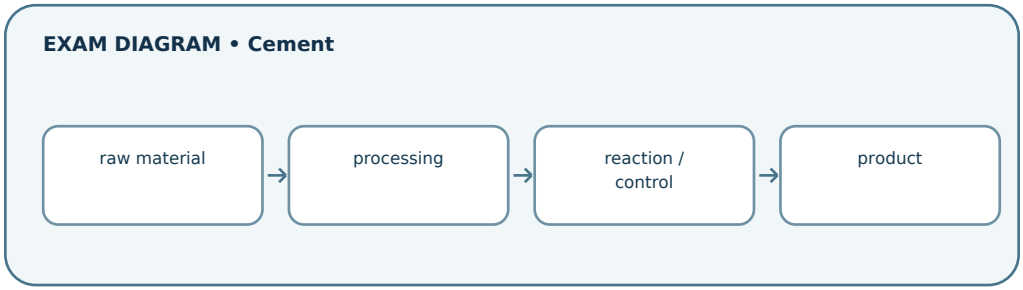
Answer: A. Gypsum regulates the setting of cement and prevents excessively rapid setting.

Question 2. In the diagram, which labelled step comes immediately after “raw material”?

A. processing
B. reaction / control
C. product
D. raw material

Answer: A. The schematic is arranged in sequence: raw material → processing → reaction / control → product.

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.



3.16 Sugar manufacture

Core concept

Sugar manufacture processes sugar-containing plant material, especially sugarcane, to obtain crystallized sugar. Operations include extraction of juice, clarification, concentration by evaporation, crystallization and separation of crystals. Explain why impurities are removed and why evaporation is used before crystallization.

Remember

Extraction → clarification → evaporation → crystallization

Sugar manufacture	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. List the main steps in sugar manufacture.

B. Why is juice concentrated before crystallization?

3.16 • Exam diagram & MCQ practice

Key exam idea: Sugar manufacture involves extracting juice, clarification, concentration, crystallization and separation of crystals.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. Why is sugar juice concentrated before crystallization?

A. to increase dissolved sugar concentration
B. to remove all sugar
C. to convert sucrose into cellulose
D. to lower the boiling point to zero

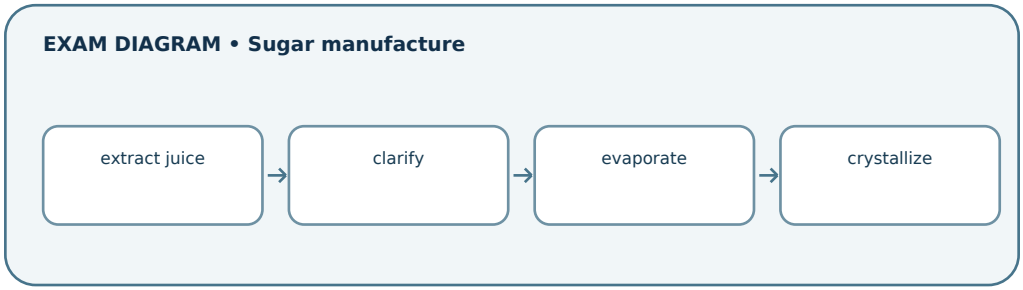
Answer: A. Evaporation removes water, making the solution concentrated enough for sugar crystals to form.

Question 2. In the diagram, which labelled step comes immediately after “extract juice”?

A. clarify
B. evaporate
C. crystallize
D. extract juice

Answer: A. The schematic is arranged in sequence: extract juice → clarify → evaporate → crystallize.

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.



3.17 Paper and pulp

Core concept

Paper production begins by separating cellulose fibres from plant material to make pulp. Pulp is processed, cleaned and formed into a sheet; water is removed and the sheet is pressed and dried. The textbook compares industrial production and material sources. Environmental issues include water use, energy use and forest-resource management.

Remember

Plant fibres → pulp → sheet formation → pressing/drying

Paper and pulp	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. Outline the stages of paper manufacture.

B. What is pulp, and why is it needed in paper production?

3.17 • Exam diagram & MCQ practice

Key exam idea: Pulp is a mass of separated cellulose fibres; it is cleaned, formed into sheets, pressed and dried to make paper.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. What is pulp in paper manufacture?

A. a suspension of separated cellulose fibres
B. a solution of pure glucose
C. molten silica
D. a synthetic monomer

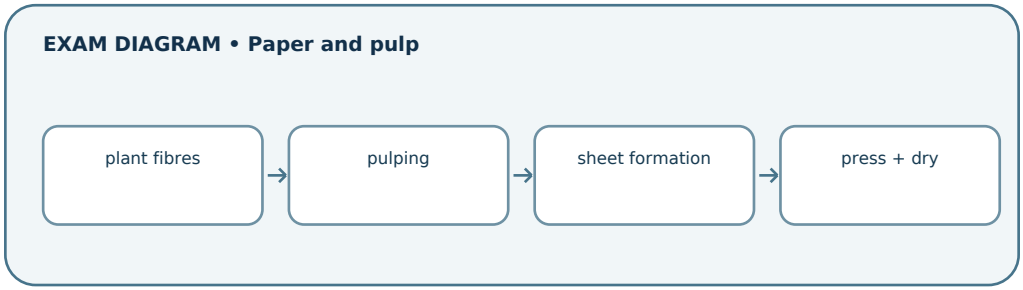
Answer: A. Pulp provides the cellulose fibres that interlock to form a paper sheet.

Question 2. In the diagram, which labelled step comes immediately after “plant fibres”?

A. pulping
B. sheet formation
C. press + dry
D. plant fibres

Answer: A. The schematic is arranged in sequence: plant fibres → pulping → sheet formation → press + dry.

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.



3.18 Tanning and food preservation

Core concept

Tanning converts animal skin or hide into leather by treating it so that it becomes more durable and resistant to decay. The unit also compares local and industrial food preservation methods. Preservation slows spoilage by controlling microorganisms or conditions such as moisture and temperature. State the principle behind each method and compare its advantages and limitations.

Remember

Tanning: hide/skin → treated leather

Tanning and food preservation	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. Explain the purpose of tanning and name two uses of leather.

B. Compare two food-preservation methods described in the unit.

3.18 • Exam diagram & MCQ practice

Key exam idea: Tanning stabilizes animal hides by treating collagen so that leather becomes more durable and resistant to decay.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. The main purpose of tanning is to:

A. convert raw hide into durable leather
B. extract sucrose from sugarcane
C. make paper fibres
D. produce ammonia

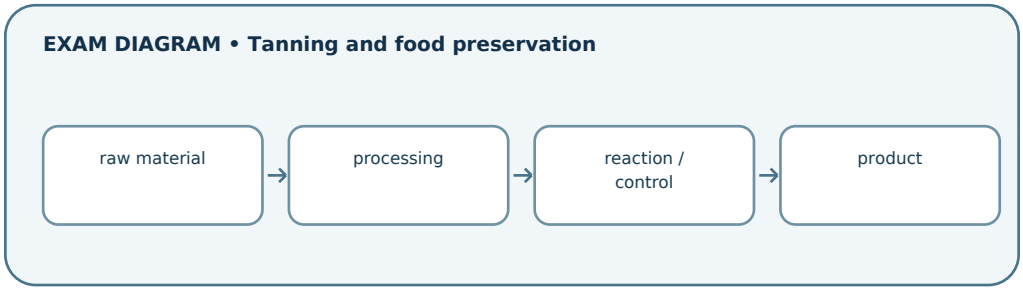
Answer: A. Tanning chemically stabilizes the hide's collagen structure and improves resistance to deterioration.

Question 2. In the diagram, which labelled step comes immediately after “raw material”?

A. processing
B. reaction / control
C. product
D. raw material

Answer: A. The schematic is arranged in sequence: raw material → processing → reaction / control → product.

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.



3.19 Ethanol, soap and detergent

Core concept

Ethanol can be manufactured by fermentation of sugars by microorganisms under suitable conditions, followed by separation and purification. Soap manufacture involves saponification of fats or oils with alkali. Detergents are cleaning agents with surfactant action and can differ from soaps in composition and behavior in hard water. Learn feedstocks, key reaction and product uses.

Remember

Fermentation: sugar → ethanol + CO■; saponification: fat/oil + alkali → soap + glycerol

Ethanol, soap and detergent	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. Write the fermentation equation for glucose.

B. Distinguish soap from detergent and describe how soap is manufactured.

3.19 • Exam diagram & MCQ practice

Key exam idea: Ethanol can be produced by fermentation of sugars; soap is made by saponification of fats or oils with alkali.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. Which reaction produces soap from fats or oils?

A. saponification
B. polymerization of ethene
C. neutralization of HCl only
D. electrolysis of water

Answer: A. Saponification hydrolyses triglycerides with alkali to form glycerol and fatty-acid salts (soap).

Question 2. In the diagram, which labelled step comes immediately after “raw material”?

A. processing
B. reaction / control
C. product
D. raw material

Answer: A. The schematic is arranged in sequence: raw material → processing → reaction / control → product.

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.

